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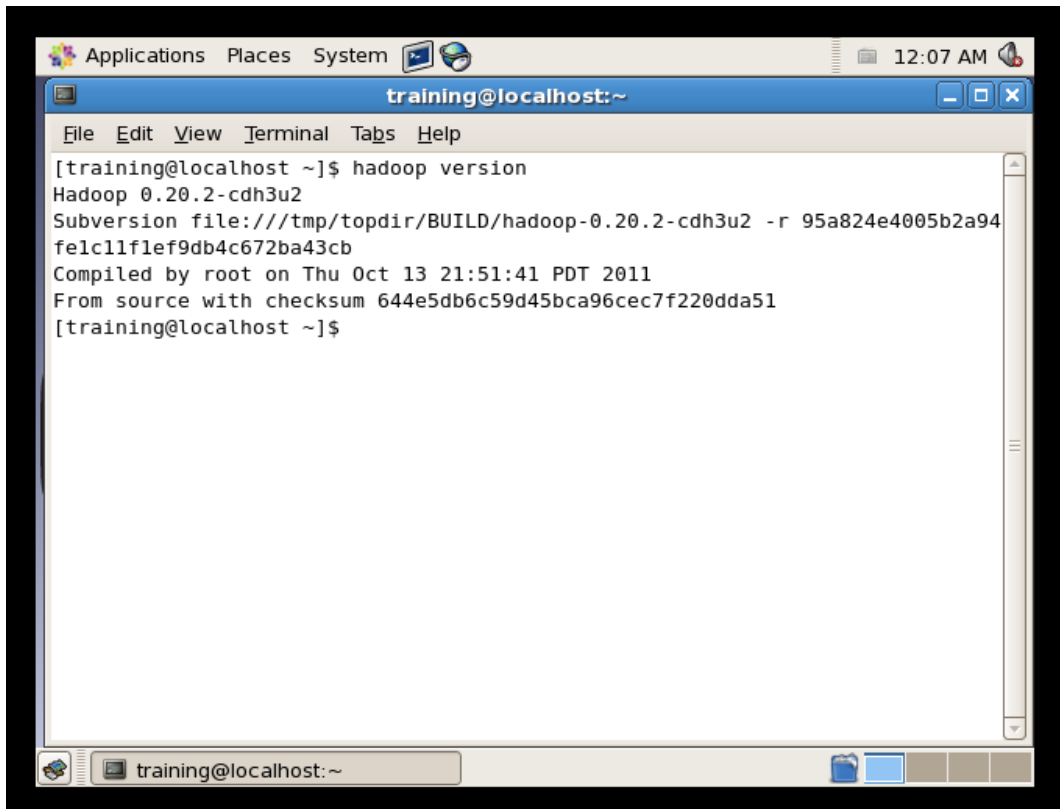
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EXPERIMENT 2

Aim: Study of Hadoop Ecosystem and HDFS Commands.

Commands:

1. **hadoop version**



```
Applications Places System 12:07 AM
training@localhost:~
File Edit View Terminal Tabs Help
[training@localhost ~]$ hadoop version
Hadoop 0.20.2-cdh3u2
Subversion file:///tmp/topdir/BUILD/hadoop-0.20.2-cdh3u2 -r 95a824e4005b2a94
felc11f1ef9db4c672ba43cb
Compiled by root on Thu Oct 13 21:51:41 PDT 2011
From source with checksum 644e5db6c59d45bca96cec7f220dda51
[training@localhost ~]$
```

2. **hadoop fs-ls**



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The screenshot shows a terminal window titled 'training@localhost:~' with a menu bar (File, Edit, View, Terminal, Tabs, Help) and a status bar (training@localhost:~). The terminal displays the output of a command, likely 'ls -l', showing file permissions, ownership, size, and timestamps for several files in the /user/trainin directory. A yellow tooltip is visible over the first line of the output, stating 'Click here to hide all windows and show the desktop.' The files listed are g/testout, g/textfile.txt, g/tusharlininput, g/tusharloutput, g/tusharloutput9, g/tusharinput, g/tusharoutput, g/tusharoutput1, g/tusharoutput2, and g/user.

File	Permissions	Owner	Group	Size	Timestamp	Path
g/testout	-rw-r--r--	1	training	super	group	24 2016-02-12 00:40 /user/trainin
g/textfile.txt	drwxr-xr-x	-	training	super	group	0 2020-11-07 00:08 /user/trainin
g/tusharlininput	drwxr-xr-x	-	training	super	group	0 2020-11-07 00:13 /user/trainin
g/tusharloutput	drwxr-xr-x	-	training	super	group	0 2020-11-07 00:24 /user/trainin
g/tusharloutput9	drwxr-xr-x	-	training	super	group	0 2020-11-06 22:17 /user/trainin
g/tusharinput	drwxr-xr-x	-	training	super	group	0 2020-11-06 22:12 /user/trainin
g/tusharoutput	drwxr-xr-x	-	training	super	group	0 2020-11-06 22:14 /user/trainin
g/tusharoutput1	drwxr-xr-x	-	training	super	group	0 2020-11-06 22:17 /user/trainin
g/tusharoutput2	drwxr-xr-x	-	training	super	group	0 2015-09-12 02:33 /user/trainin
g/user	-rw-r--r--	1	training	super	group	59 2015-10-10 00:56 /user/trainin

3. hadoop fs



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The screenshot shows a terminal window titled "training@localhost:~". The window has a menu bar with "File", "Edit", "View", "Terminal", "Tabs", and "Help". The terminal content lists various Hadoop fs commands, including [-du <path>], [-dus <path>], [-count[-q] <path>], [-mv <src> <dst>], [-cp <src> <dst>], [-rm [-skipTrash] <path>], [-rmr [-skipTrash] <path>], [-expunge], [-put <localsrc> ... <dst>], [-copyFromLocal <localsrc> ... <dst>], [-moveFromLocal <localsrc> ... <dst>], [-get [-ignoreCrc] [-crc] <src> <localdst>], [-getmerge <src> <localdst> [addnl]], [-cat <src>], [-text <src>], [-copyToLocal [-ignoreCrc] [-crc] <src> <localdst>], [-moveToLocal [-crc] <src> <localdst>], [-mkdir <path>], [-setrep [-R] [-w] <rep> <path/file>], [-touchz <path>], [-test [-ezd] <path>], and [-training@localhost:~]h>]. The prompt [-training@localhost:~]h> is highlighted with a yellow box. The terminal window has a status bar at the bottom showing "training@localhost:~" and some icons.

```
[-du <path>]
[-dus <path>]
[-count[-q] <path>]
[-mv <src> <dst>]
[-cp <src> <dst>]
[-rm [-skipTrash] <path>]
[-rmr [-skipTrash] <path>]
[-expunge]
[-put <localsrc> ... <dst>]
[-copyFromLocal <localsrc> ... <dst>]
[-moveFromLocal <localsrc> ... <dst>]
[-get [-ignoreCrc] [-crc] <src> <localdst>]
[-getmerge <src> <localdst> [addnl]]
[-cat <src>]
[-text <src>]
[-copyToLocal [-ignoreCrc] [-crc] <src> <localdst>]
[-moveToLocal [-crc] <src> <localdst>]
[-mkdir <path>]
[-setrep [-R] [-w] <rep> <path/file>]
[-touchz <path>]
[-test [-ezd] <path>]
[-training@localhost:~]h>
```

4. hadoop fs -df hdfs:/



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```
Applications Places System 12:17 AM
training@localhost:~
File Edit View Terminal Tabs Help
[-chgrp [-R] GROUP PATH...]
[-help [cmd]]

Generic options supported are
-conf <configuration file>      specify an application configuration file
-D <property=value>             use value for given property
-fs <local|namenode:port>       specify a namenode
-jt <local|jobtracker:port>     specify a job tracker
-files <comma separated list of files> specify comma separated files to be copied to the map reduce cluster
-libjars <comma separated list of jars> specify comma separated jar files to include in the classpath.
-archives <comma separated list of archives> specify comma separated archives to be unarchived on the compute machines.

The general command line syntax is
bin/hadoop command [genericOptions] [commandOptions]

[training@localhost ~]$ hadoop fs -df hdfs:/
Filesystem      Size      Used    Avail    Use%
hdfs:/          18611908608  99086336 12415631360  0%
[training@localhost ~]$
```

5. `hadoop fs -count hdfs:/`



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```
Applications Places System 12:19 AM
training@localhost:~
File Edit View Terminal Tabs Help

Generic options supported are
-conf <configuration file>      specify an application configuration file
-D <property=value>             use value for given property
-fs <local|namenode:port>       specify a namenode
-jt <local|jobtracker:port>     specify a job tracker
-files <comma separated list of files> specify comma separated files to be
copied to the map reduce cluster
-libjars <comma separated list of jars> specify comma separated jar files
to include in the classpath.
-archives <comma separated list of archives> specify comma separated arch
ives to be unarchived on the compute machines.

The general command line syntax is
bin/hadoop command [genericOptions] [commandOptions]

[training@localhost ~]$ hadoop fs -df hdfs:/
Filesystem      Size      Used      Avail      Use%
hdfs:/          18611908608  99086336  12415631360  0%
[training@localhost ~]$ hadoop fs -count hdfs:/
          419          448          94581956 hdfs://localhost/
[training@localhost ~]$
```

6. hadoop fs -ls user



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```
Applications Places System 12:20 AM
training@localhost:~
File Edit View Terminal Tabs Help
- fs <local|namenode:port>      specify a namenode
- jt <local|jobtracker:port>    specify a job tracker
- files <comma separated list of files> specify comma separated files to be copied to the map reduce cluster
- libjars <comma separated list of jars> specify comma separated jar files to include in the classpath.
- archives <comma separated list of archives> specify comma separated archives to be unarchived on the compute machines.

The general command line syntax is
bin/hadoop command [genericOptions] [commandOptions]

[training@localhost ~]$ hadoop fs -df hdfs:/
Filesystem      Size      Used      Avail      Use%
hdfs:/          18611908608  99086336  12415631360  0%
[training@localhost ~]$ hadoop fs -count hdfs:/
      419      448      94581956 hdfs://localhost/
[training@localhost ~]$ hadoop fs -ls user
Found 1 items
drwxr-xr-x  - training supergroup      0 2015-09-12 02:33 /user/training
g/user/training
[training@localhost ~]$
```

7. hadoop balancer



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```
bin/hadoop command [genericOptions] [commandOptions]

[training@localhost ~]$ hadoop fs -df hdfs:/
Filesystem      Size      Used      Avail      Use%
hdfs:/          18611908608  99086336  12415631360  0%
[training@localhost ~]$ hadoop fs -count hdfs:/
      419      448      94581956 hdfs://localhost/
[training@localhost ~]$ hadoop fs -ls user
Found 1 items
drwxr-xr-x  - training supergroup          0 2015-09-12 02:33 /user/training
g/user/training
[training@localhost ~]$ hadoop balancer
Time Stamp      Iteration#  Bytes Already Moved  Bytes Left To Move
Bytes Being Moved
25/02/03 00:20:46 INFO net.NetworkTopology: Adding a new node: /default-rack
/127.0.0.1:50010
25/02/03 00:20:46 INFO balancer.Balancer: 0 over utilized nodes:
25/02/03 00:20:46 INFO balancer.Balancer: 1 under utilized nodes: 127.0.0.1
:50010
The cluster is balanced. Exiting...
Balancing took 133.0 milliseconds
[training@localhost ~] training@localhost:~
```

8. `hadoop fs -mkdir /Pikachu`

```
mkdir: cannot create directory /Qulthum: File exists
[training@localhost ~]$ hadoop fs -mkdir /Pikachu
[training@localhost ~]$
```

9. `hadoop fs -ls/`



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```
[training@localhost ~]$ hadoop fs -mkdir /Pikachu
[training@localhost ~]$ hadoop fs -ls /
Found 16 items
drwxr-xr-x - training supergroup 0 2025-02-03 00:49 /Pikachu
drwxr-xr-x - training supergroup 0 2025-02-03 00:39 /Qulthum
drwxr-xr-x - training supergroup 0 2024-08-22 04:30 /Rishi
drwxr-xr-x - training supergroup 0 2024-10-03 04:04 /RishiShah
drwxr-xr-x - training supergroup 0 2024-08-22 03:56 /STestFile1
drwxr-xr-x - hbase supergroup 0 2024-08-22 03:19 /hbase
drwxr-xr-x - training supergroup 0 2015-10-10 23:02 /home
drwxr-xr-x - training supergroup 0 2024-10-03 04:37 /pigresult1
drwxr-xr-x - training supergroup 0 2024-11-07 02:09 /qulthum
drwxr-xr-x - training supergroup 0 2025-02-03 00:20 /system
drwxrwxrwx - hue supergroup 0 2024-11-07 01:58 /tmp
drwxr-xr-x - training supergroup 0 2016-01-18 08:25 /training
drwxr-xr-x - hue supergroup 0 2021-10-13 03:18 /user
drwxr-xr-x - mapred supergroup 0 2021-10-13 03:18 /var
drwxr-xr-x - training supergroup 0 2024-11-07 02:10 /vidhi
drwxr-xr-x - training supergroup 0 2024-11-07 02:14 /vps
[training@localhost ~]$
```

10. `hadoop fs -mkdir /Pikachu/pikapika.txt`

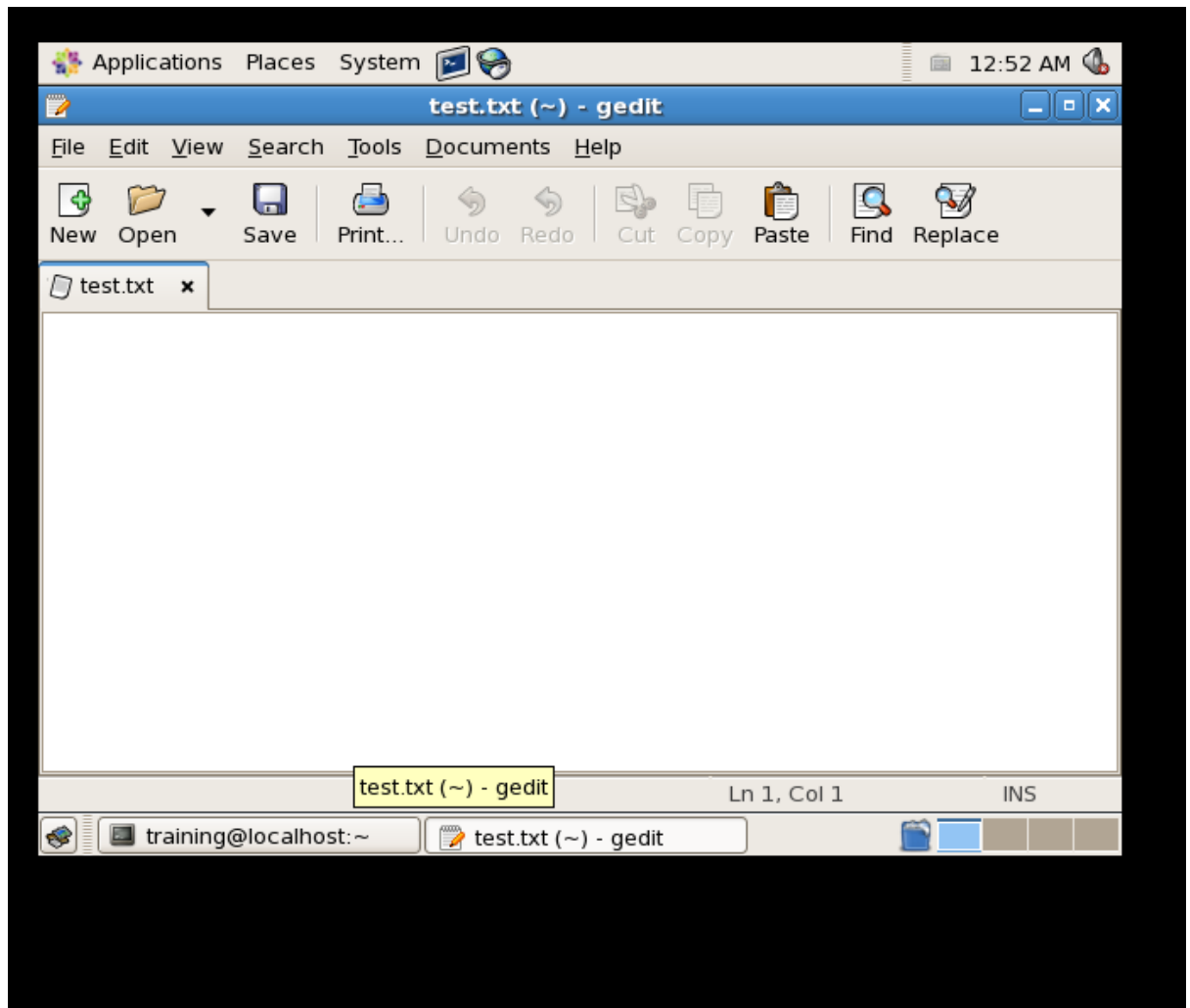
```
drwxr-xr-x - training supergroup 0 2024-11-07 02:10 /vidhi
drwxr-xr-x - training supergroup 0 2024-11-07 02:14 /vps
[training@localhost ~]$ hadoop fs -mkdir /Pikachu/pikapika.txt
[training@localhost ~]$
```

11. `gedit test.txt`

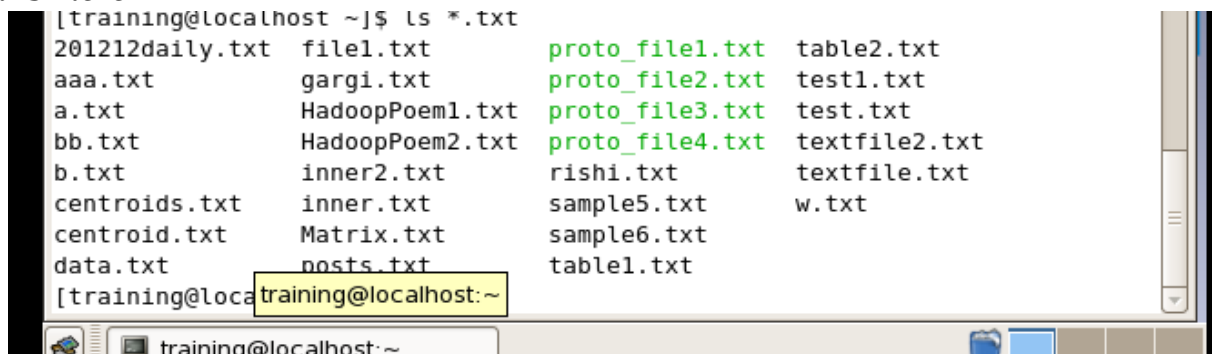


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12. ls *.txt



13.hadoop fs -put



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```
data.txt      posts.txt      table1.txt
[training@localhost ~]$ hadoop fs -put test.txt/Pikachu/pikapika.txt
Usage: java FsShell [-put <localsrc> ... <dst>]
[training@localhost ~]$
```

14. `hadoop fs -get /Pikachu/pikapika.txt/test.txt`

```
cat: test.txt: No such file or directory
[training@localhost ~]$ hadoop fs -get test.txt/Pikachu/pikapika.txt
Usage: java FsShell [-get [-ignoreCrc] [-crc] <src> <localdst>]
[training@localhost ~]$ cat test.txt
```

15. `cat test.txt`

```
[training@localhost ~]$ cat test.txt
hihihi
[training@localhost ~]$ rm test.txt
```

16. `rm test.txt`

```
[training@localhost ~]$ rm test.txt
rm: remove regular file `test.txt'?
```

Conclusion:

Thus, in this experiment we got familiar with the hadoop environment and how to execute hadoop commands. Using this we were successful in implementing the above mentioned commands and received favourable result regarding the same. We studied how hadoop helps in storing and processing of big data and practically experienced the significance and benefits that come with using such as environment.